Krishiv Singhal

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Education

Carnegie Mellon University *2024–Dec 2027*

Mathematics and Logic *United States*

- Notable coursework: Advanced NLP (PhD), Machine Learning (PhD), Cooperative AI (Graduate), Modal Logic (Graduate), Formal Logic, Probability Theory, Linear Algebra, Analysis.
- Studied abroad at the University of Sydney; completed advanced coursework in maths on a full merit scholarship; awarded the Dalyell Scholarship and Chancellor's Award.

United World College, Singapore *2017–2024*

International Baccalaureate *Singapore*

- Perfect IB Score: 45/45 (top 0.5% worldwide); ranked first in class for thesis on the Curry–Howard correspondence and intuitionism.

Experience

Quantitative Research Intern *Jun–Aug 2026*

Stealth Systematic Macro Fund; led by former Goldman Sachs Managing Director *Hong Kong*

- Built Bayesian state-estimation engine that classified global equity indices and stocks into market regimes using Gaussian-kernel geometry; integrated into the fund's research stack.
- Developed co-activity and flow measures tracking probability mass between regimes, isolating churn conditions.

Modal Logic and Reinforcement Learning *Mar 2026–Present*

Research, Carnegie Mellon University

- Formulated novel probabilistic semantics for propositional dynamic logic by relating programs to Markov decision processes; researching its use as a framework for reinforcement learning.

Computational Linguistics Research Group *Sep 2024–Feb 2025*

Carnegie Mellon University

- Analyzed human and AI generated writing through eigendecomposition of stochastic matrices; found statistically significant variations in Markov Chain mixing time(s).

Projects

Llama-Style Language Model *PyTorch*

- Implemented a 25M-parameter Llama-style model from scratch in PyTorch, including BPE, rotary embeddings, grouped-query attention, RMSNorm, SwiGLU, and AdamW; trained it on 1,941 books and essays (212M tokens), upsampling Russell, Hofstadter, Wilde, and my own essays.

Differentiable Kripke Semantics for Modal Model Search *PyTorch, NumPy*

- Implemented differentiable fuzzy Kripke semantics and framed modal satisfiability as a minimax optimisation problem. Developing framework using modal logic as a regulariser for RL policy learning.

Bitewise Labs — Bites Suite *Swift, SwiftUI, Metal*

- Founded an iOS studio with three shipped cultural-education apps (Literature, Art and Architecture Bites) with thousands of impressions and consistent revenue. Built automated video-generation and 3D monument-rendering content pipelines.

FO^2 Presentation

- Researched, prepared and delivered a talk to faculty and members of the Australasian logic seminar on the decision problem for the binary fragment of first-order logic.

Paraconsistent Logic, Monash University *Lean 4, Isabelle*

- Formalized components of Weber's paraconsistent logic and set theory in Isabelle and Lean 4 whilst in high school.

Awards

Chancellor's Award, University of Sydney · **UWC Global Academy,** University of Chicago · **Canadian Senior Maths Contest,** International Honour Roll, global top 1% · **AIME,** Qualifier, global top 2.5% · **Cayley Maths Competition,** International Honour Roll, top 2% of >17,500.

Skills and Interests

Python (NumPy, pandas, PyTorch), Swift (SwiftUI, Metal), Lean 4, Isabelle, TypeScript, SQL, LaTeX.

Interests: Music production (composed, produced, released 5 albums on major streaming platforms; remix favourite songs on YouTube); golf; poker.